# Mario Torres Danta

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Seville, Spain

## Education

**B.Sc. in Mathematics** *2023 – 2027 (Expected)*

*University of Seville (US), Seville, Spain*

**Current GPA: 8.74 / 10.0** (Top 5% of cohort; based on 22 graded courses)

**Bachelor's thesis (in progress):** Algebraic K-theory. Supervisor: Prof. Ramón J. Flores.

## Academic Honors and Distinctions

**Collaboration Grant (*Beca de Colaboración*), Spanish Government** *2026 – 2027*

Competitive national research-initiation grant for top final-year students, held in a university department.

Awarded **"Matrícula de Honor" (Highest Distinction)** in several courses, such as:

- **General Topology (10.0)**
- **Mathematical Logic (10.0)**
- **Measure Theory & Integration (10.0)**
- **Mathematical Optimization (9.2)**
- **Numerical Analysis I (9.5)**
- **Ordinary Differential Equations (9.4)**
- **Probability Theory (9.4)**

*Additional "Sobresaliente" (A-Grade) in courses such as Local Geometry of Surfaces (10.0), Algebraic Structures, Complex Analysis, Linear Algebra II, Discrete Mathematics.*

## Research Experience

**Visiting Student & Undergraduate Researcher** *Aug 2026 – Sep 2026*

*University of Copenhagen, Denmark*

**Project:** Mapping class groups of finite abelian branched covers of surfaces. Identified the liftable and symmetric mapping class groups with mixed braid groups, described the lifts explicitly via chain twists and multi-Dehn twists, and derived homological stability results. Resulted in a manuscript in preparation (see below).

**Supervisor:** Prof. Nathalie Wahl

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**FUSRP Undergraduate Researcher** *Jun 2026 – Jul 2026*

*The Fields Institute for Research in Mathematical Sciences, Toronto, Canada*

**Project:** Laurent symmetric functions, which extend classical symmetric functions to the rational representations of general linear groups. Studied their ring structure, constructed orthonormal bases for the Hall inner product, and applied them to representations of $\mathrm{GL}_n$ and $\mathrm{GL}_\infty$. Resulted in a preprint (see below).

**Supervisors:** Prof. Alistair Savage and Dr. Yaolong Shen

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**Undergraduate Researcher (Departmental Collaborator)** *Sep 2025 – Jun 2026*

*Department of Geometry and Topology, University of Seville*

**Project:** Advanced topics in Topology and Group Theory. A guided study complementing the standard undergraduate curriculum.

**Supervisor:** Prof. Ramón J. Flores

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**Undergraduate Researcher (Departmental Collaborator)** *Sep 2024 – Jun 2025*

*Department of Algebra, University of Seville*

**Project:** Classification of homographies and homologies in real ($\mathbb{P}^n(\mathbb{R})$) and complex ($\mathbb{P}^n(\mathbb{C})$) projective spaces.

**Supervisor:** Prof. Francisco Castro Jiménez

## Publications and Manuscripts

H. Fernandez, E. Kleinwort, T. Li and **M. Torres-Danta**, *Laurent symmetric functions*, preprint (2026). arXiv:2609.15690.

**M. Torres-Danta**, *The symmetric mapping class group of finite abelian branched covers*, manuscript in preparation (2026).

## Talks

***Laurent symmetric functions*** *Jul 2026*

Final presentation, Fields Undergraduate Summer Research Program, Fields Institute, Toronto.

***Mapping class groups of finite abelian branched covers*** *Sep 2026*

Department of Mathematical Sciences, University of Copenhagen.

## Research Interests

Algebraic Topology (Homology, Homotopy Theory, Homological Stability) · Geometric Topology (Mapping Class Groups, Moduli Spaces) · Algebraic K-Theory · Group Theory · Category Theory

## Academic Service

**Reviewer, zbMATH Open** *Sep 2026 – Present*

Writing reviews of recent research articles for the zbMATH Open database.

## Academic Engagement

**ByMat Conference (Bringing Young Mathematicians Together)** *Nov 2025*

*University of Seville / RSME (Royal Spanish Mathematical Society)*

**Role: Attendee.** Highlights: plenary lectures by **Prof. Marithania Silvero** (Knot Theory) and **Prof. Alberto Rodríguez** (Differential Geometry/Topology).

## Skills

- **Languages:** Spanish (Native), English (C1), French (B2)
- **Computing: Python** (*Santander Open Academy* certificate), **MATLAB**, **LaTeX**, **GitHub**

## References


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**Prof. Alistair Savage**
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**Prof. Ramón J. Flores**
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